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DESIGN THINKING LABORATORY REPORT

SCIENCE UNLOCKED

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DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING



CERTIFICATE

Certified that the Design thinking Laboratory work titled '*Science Unlocked*' is carried out by **Pratham Nilange (1RV23IS091), Parthav Pratim M (1RV23IS081), Sadhu Venkata Hemanth Sai(1RV23IS100), Tanikonda Sai Karan(1RV23IS129)** in partial fulfilment for the requirement of degree of **Bachelor of Engineering in Information Science and Engineering** of the Visvesvaraya Technological University, Belagavi during the year 2024-2025. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the report.

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Name - USN

ABSTRACT

Science Unlocked is an innovative educational project that addresses the need for a chronological understanding of science to foster an intuitive appreciation of its evolution. Targeting students, educators, and science enthusiasts, the platform serves as an interactive resource showcasing the progression of scientific milestones across disciplines such as physics, and chemistry. By integrating timelines and linking discoveries across time, the platform highlights how scientific thought has evolved. The project taps into the growing EdTech market, projected to expand significantly. Despite the availability of general educational platforms, few focus on contextualizing scientific breakthroughs in chronological order, marking this as a unique and valuable proposition. It provides an opportunity for learners themselves to discover topics as it was done historically.

Despite significant advancements in EdTech, most platforms emphasize concept delivery without showcasing the historical and contextual development of scientific theories. This gap limits learners' understanding of how scientific knowledge builds upon past discoveries. The project addresses this unmet need by offering a timeline-based approach that integrates key milestones, interactive features, and multimedia resources. The platform design was informed by thorough competitor analysis and user research, identifying a demand for intuitive and engaging tools for learning science. The methodology builds on interactive web technologies like HTML5, JavaScript, and CSS, ensuring cross-platform usability. Wireframes and prototypes of features like clickable timelines, cross-disciplinary links, and multimedia expansions were tested iteratively to refine the user experience.

The project followed a structured methodology, beginning with user research to define the needs of target users. Content curation involves mapping key scientific milestones and organizing them into interactive timelines. The platform's user interface (UI) and user experience (UX) were developed using design tools like Figma, with technical development employing front-end and back-end technologies for dynamic interactions and database integration. Constraints included balancing user engagement with factual accuracy, which was ensured through expert review and fact-checking. The platform was subjected to usability tests and beta trials with target users, allowing iterative refinements before launch. Science Unlocked exemplifies how a methodical approach to content, design, and technology can create an accessible and engaging educational resource that inspires curiosity and a deeper appreciation for science.

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2. Empathy

2.1 Client Details

Primary Clients:

Students

- **Segment:** Middle school, high school, and undergraduate students.
- **Need:** To gain a better understanding of science concepts and their historical evolution, enhance academic performance, and develop an intuitive grasp of interconnected scientific theories.
- **How Science Unlocked Meets Their Needs:**
 - Interactive timelines and engaging multimedia content for self-paced learning.
 - Simplified explanations of complex theories with historical context.

Schools and Educational Institutions

- **Segment:** Public and private schools, coaching centers, and curriculum developers.
- **Need:** To integrate innovative learning tools into science curricula and improve student engagement with scientific concepts.
- **How Science Unlocked Meets Their Needs:**
 - Ready-to-use teaching aids like timelines, quizzes, and videos.
 - Tools to support experiential and project-based learning.

Educators

- **Segment:** School teachers, college professors, and private tutors.
- **Need:** Access to teaching resources that connect theoretical knowledge with historical scientific developments to improve classroom engagement.
- **How Science Unlocked Meets Their Needs:**
 - Reliable and curated content that complements lessons.
 - Visual and interactive tools to enhance science instruction.

Secondary Clients :

Science Enthusiasts

- **Segment:** Hobbyists, independent learners, and lifelong learners interested in science.
- **Need:** A resource to explore scientific breakthroughs, cross-disciplinary connections, and historical insights.
- **How Science Unlocked Meets Their Needs:**
 - A user-friendly platform to explore science in a chronological and engaging manner.
 - Unique content highlighting lesser-known scientific developments.

Libraries and Learning Centers

- **Segment:** Public libraries, online knowledge hubs, and STEM-focused learning centers.
- **Need:** Educational platforms that encourage independent study and promote STEM education.
- **How Science Unlocked Meets Their Needs:**
 - A resource to complement existing materials for independent learners.
 - A scalable digital platform that can be accessed remotely.

Science Outreach Programs

- **Segment:** Non-profits, museums, and science festivals promoting STEM education.
- **Need:** Tools to engage diverse audiences in understanding science in a fun and approachable way.
- **How Science Unlocked Meets Their Needs:**
 - Engaging timelines and interactive demonstrations for outreach events.
 - Content designed for all age groups and educational backgrounds.

Client Needs

- High-quality, engaging science education materials.
- Structured learning paths that build upon previous knowledge.
- Resources that simplify lesson planning and provide hands-on experiences.

Benefits to Clients

- **Comprehensive Kits:** Each kit is designed to minimize preparation time for educators and parents, providing all necessary materials in one package.
- **Flexibility:** Clients can choose between full-year curricula or individual monthly kits, allowing customization based on their educational goals.
- **Engagement:** The hands-on nature of the curriculum keeps students interested and motivated to learn science in a fun way.

2.2 Need Analysis with Evidence

A needs analysis systematically identifies gaps between current and desired educational outcomes. It involves gathering data on audience knowledge and skills to inform project design. This process helps clarify the necessity for a project, ensuring alignment with strategic goals and stakeholder needs, ultimately guiding effective resource allocation and project implementation .

2.2.1 Questionnaire and Analysis

The questionnaire serves as a key tool in the needs analysis process, collecting qualitative and quantitative data from stakeholders. This data is analyzed to identify skill gaps, assess current practices, and determine specific educational needs. The results inform curriculum development and targeted interventions, ensuring that educational offerings are relevant and effective .

Will you understand a topic more intuitively if you understood how the topic was derived ? i.e., understanding the background story of the topics

47 responses

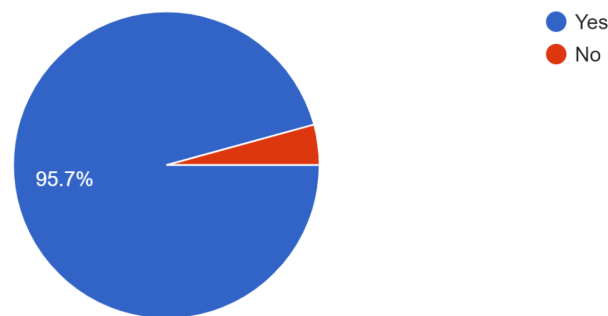


Fig. 1: Response 1

Most individuals find topics more intuitive and comprehensible when they understand the background or context in which those topics were derived. Context deepens understanding and developmental context of a topic helps learners connect abstract concepts to real-world events or thought processes.

What types of content would you find most engaging on an educational website?

48 responses

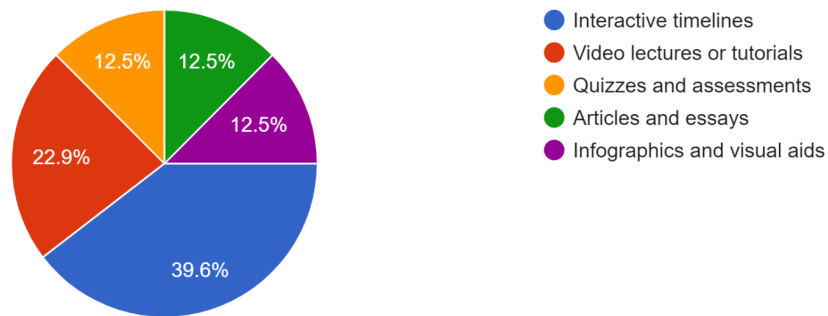


Fig. 2: Response 2

The **inference** from this result is that **timelines** are the most engaging type of content for the target audience, followed by video lectures. This aligns closely with the goals of your project, **Science Unlocked**. It indicates that users value chronological representations of information as an intuitive way to learn. This validates your platform's primary focus on interactive timelines to showcase the evolution of scientific thought.

How important is it for you to learn science in chronological order?

47 responses

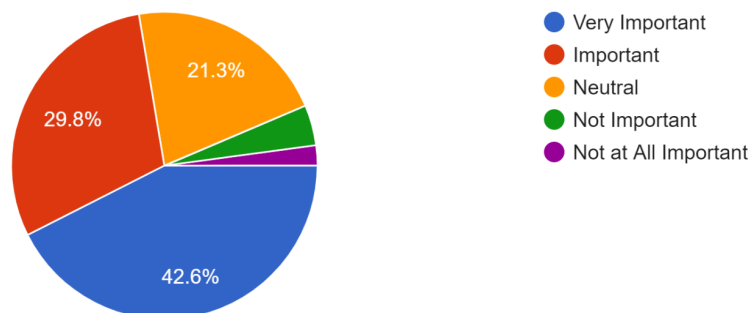


Fig. 3: Response 3

2.2.2 Other Materials

Effectiveness of Chronological and Immersive Content:

- Research confirms that chronological narratives and context-driven education help students connect historical discoveries to modern applications, fostering critical thinking and retention (src: Carnegie Mellon University)

Demand for Conceptual Understanding Over Rote Learning:

- Interactive content such as timelines, simulations, and real-world examples enhances engagement and comprehension, as opposed to traditional text-heavy materials
(src: skillademia)

Interactive Learning and Active Engagement:

- Studies show that active learning methods, like combining physical and virtual interactions, significantly improve students' conceptual understanding and ability to apply knowledge in real-world contexts.
- 82% of Harvard professors express interest in integrating digital tools into their teaching to enhance student engagement

Enhanced Engagement Through Multimodal Learning

- Digital timelines incorporate text, images, videos, games, and audio to provide dynamic learning experiences. These elements increase motivation and engagement by making abstract scientific concepts more accessible and relatable to learners (src: U.S. Department of Education)

Improved Knowledge Construction

- The use of digital timelines allows students to construct knowledge interactively by linking events, contexts, and multimedia resources. This enables a deeper understanding of the interconnectedness of historical events, scientific discoveries, and cultural context. (src: U.S. Department of Education)

Historical Context in Science Education

- Digital timelines effectively situate scientific discoveries within their historical and sociocultural contexts, helping learners understand the influence of time, politics, and culture on the development of scientific thought. (src: U.S. Department of Education)

3 Define

3.1 Problem Statement

Despite the wealth of online educational resources, many learners find it difficult to comprehend the interconnectedness of scientific concepts and the chronological progression of scientific discoveries. Existing platforms often provide fragmented information, making it hard for students, educators, and science enthusiasts to see how scientific ideas have developed over time.

There is a need for a comprehensive , interactive platform that organizes scientific knowledge chronologically and intuitively, highlighting key milestones and breakthroughs across disciplines. Such a resource would help bridge the gaps in understanding the historical and conceptual context of science while accommodating diverse learning preferences.

3.2 Empathy Map:

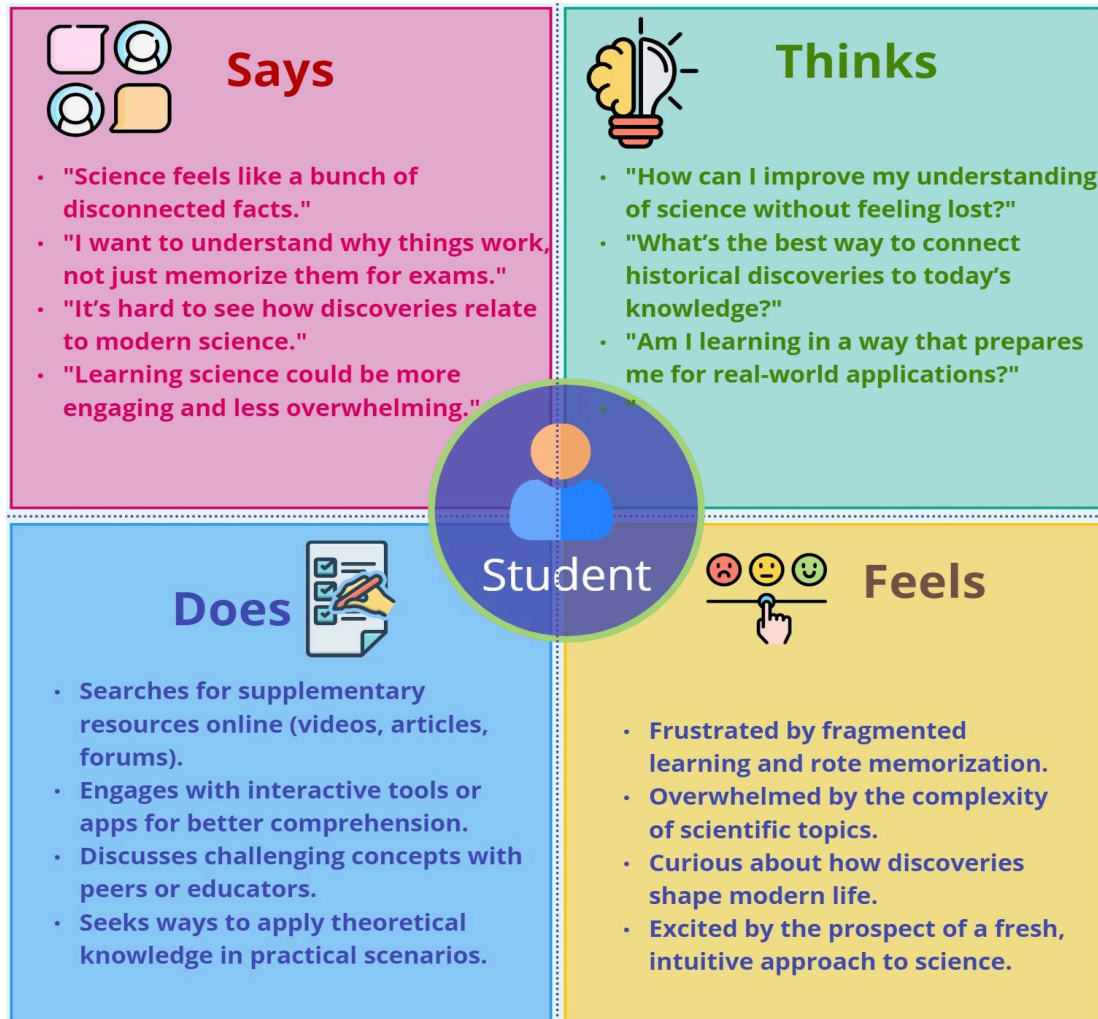


Fig. 4: Empathy Map

4. Ideate

4.1 Prime Challenges Identification

The development of *Science Unlocked* faced several key challenges:

1. **Balancing Content and Engagement:** Striking the right balance between presenting detailed scientific content and keeping it engaging for users was difficult. The platform needed to avoid overwhelming users with complex material while still delivering educational value.
2. **Ensuring Historical Accuracy:** The chronological presentation of scientific discoveries required meticulous fact-checking and validation to maintain accuracy across various scientific fields, where interpretations of milestones can vary.
3. **User Experience Design:** Creating an intuitive, user-friendly interface was crucial. The platform had to be interactive yet simple enough for users to navigate, especially for those who may not be familiar with complex interfaces.
4. **Technological Integration:** Incorporating multimedia resources, such as videos and interactive features, while ensuring platform performance and compatibility across devices and browsers was a technical challenge.
5. **Scalability and Content Updates:** As the platform expanded, ensuring it could easily incorporate new content without disrupting the user experience was essential for future growth.

5. Prototype

5.1 Technologies Used

HTML/CSS:

- Used HTML and CSS to create the structure and style of the website pages, ensuring a visually appealing and user-friendly design. HTML was utilized to structure the content and define the layout, while CSS was employed to enhance the visual presentation, including typography, colors, spacing, and responsiveness across devices.

JavaScript(Js):

- Used JavaScript to add interactivity and dynamic functionality to the website. It enables smooth animations, enhances user experience with interactive elements, and ensures the responsiveness of features like navigation and content loading. JavaScript was also used to implement data handling and improve the overall flow of the site.

Github:

- Used GitHub to manage the version control of the project, enabling efficient collaboration among team members. It allowed us to track changes, maintain a history of code updates, and seamlessly merge contributions from multiple developers. GitHub also provided a platform for continuous integration and deployment, ensuring smooth updates and deployment of the website.

5.2 Solution Developed

- Solution Overview: An innovative educational platform called *Science Unlocked*.
- Purpose: Offers a chronological understanding of scientific milestones.
- Key Features:
 - Interactive timelines.
 - Cross-disciplinary links.
 - Multimedia resources.
- Focus Areas: Highlights the evolution of scientific thought in physics, chemistry, and biology.
- Technology Used: Developed with HTML, CSS, and JavaScript for front-end.
- User Experience: Ensures a seamless and intuitive experience.
- Unique Value Proposition: Contextualizes scientific breakthroughs in their historical order, fostering a deeper understanding.
- Design & Content: Shaped by extensive user research and competitor analysis.
- Target Audience: Students, educators, and science enthusiasts.

5.3 Outputs

- An interactive web-based platform showcasing science's chronological evolution.
- A clean, user-friendly interface with clickable timelines, cross-topic links, and multimedia expansions for each scientific milestone.
- A fully functional platform tested across multiple devices to ensure responsiveness and accessibility.
- A user interface and experience designed with Figma, followed by front-end and back-end technologies for dynamic content interaction and database integration.

6. Test

6.1 Field Trials

Field trials for the platform were conducted with students, educators, and science enthusiasts. Beta testing sessions were held with a select group of users, where they navigated the platform, explored different scientific topics, and provided feedback on their experience. The trials focused on testing the usability of the interactive timelines, the effectiveness of cross-disciplinary links, and the overall flow of content delivery.

6.2 Testing Details

Testing involved a series of iterations, starting with internal usability tests to ensure the platform's features, such as the clickable timelines, multimedia content, and interactive links, functioned as expected. The user interface was tested for ease of navigation, readability, and overall engagement. Performance testing ensured that the platform worked smoothly across various devices and browsers, providing a responsive and fluid user experience. The project also went through content accuracy checks, where scientific milestones were reviewed and validated by subject matter experts.

6.2 Client Feedback

Feedback from trial users highlighted the platform's educational value, particularly in providing a historical context to scientific discoveries. Users appreciated the engaging timeline format and the way it connected different scientific topics. However, some users suggested further enhancements, such as adding more detailed descriptions for specific discoveries and incorporating additional interactive features, such as quizzes or discussion

boards and videos. The client feedback was instrumental in refining the platform's content and functionality.

Conclusion And Future Scope

Science Unlocked has effectively created an engaging platform that offers a chronological understanding of scientific milestones, filling a gap in EdTech by contextualizing how scientific knowledge has evolved. The interactive features, such as clickable timelines and cross-disciplinary links, have proven effective in helping users connect key discoveries and better understand the progression of science across various fields. Feedback from users during beta testing highlighted the value of the platform in providing a historical perspective on scientific concepts, making it an important educational tool for learners of all ages.

Looking forward, the platform can be enhanced by adding more interactive elements, such as quizzes, virtual labs, or discussion forums, to further engage users and deepen their learning experience. Expanding the content to include more niche scientific fields and recent advancements would also attract a wider audience. Future developments could incorporate AI to offer personalized learning paths, making the platform even more tailored to individual users' needs and progress.

With the growing demand for EdTech solutions, *Science Unlocked* has the potential to evolve into a comprehensive, interactive educational tool for science learners at all levels. As technology continues to evolve, incorporating augmented reality (AR) or virtual reality (VR) could take the platform to the next level, providing immersive, hands-on experiences that bring scientific concepts to life in exciting new ways.

Finally, as *Science Unlocked* continues to expand, the platform could collaborate with educational institutions and research organizations to create a more robust and globally recognized resource. This partnership could help ensure that the content remains up-to-date, scientifically accurate, and aligned with current educational standards.

Science Unlocked has the potential to become a comprehensive educational resource, and with continuous improvements and collaboration with educational institutions, it can further

transform the way science is taught and understood, fostering a deeper curiosity and appreciation for scientific knowledge.

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Images



Fig 5: Landing Page

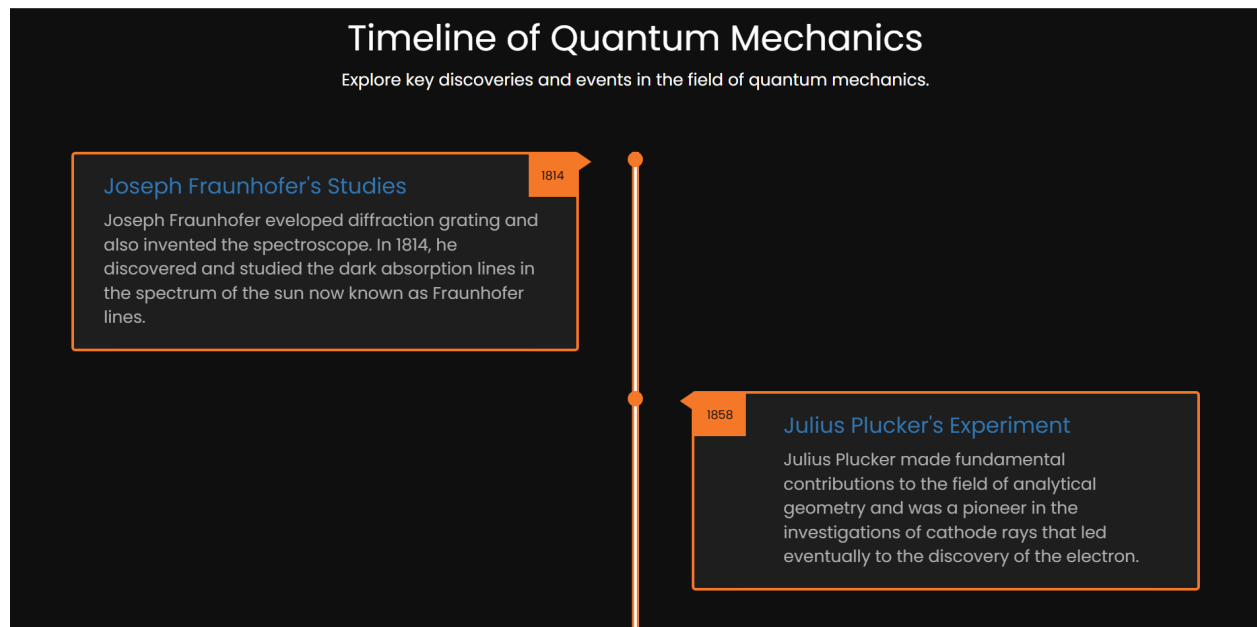


Fig 6: Timeline Page